

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

EYsJVQU5igQNaNMptsyL2A6h6QWso34MYxTriAkdQqWj

Date: 5/1/2025

Compounds: Amoxicillin, Aspirin

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Amoxylirin
Formula: C16H19N3O6
Weight: 349.34
Drug-likeness: 0.85

Activity:

Broad-spectrum antibacterial with anti-inflammatory properties

Target Analysis

Penicillin-binding proteins

Binding Affinity: 8.5

Essential for bacterial cell wall synthesis inhibition

Cyclooxygenase-1/2 (COX-1/2)

Binding Affinity: 7.3

Reduces inflammation and pain through prostaglandin inhibition

Toxicity Profile

Risk Level: Moderate

Affected Systems: Gastrointestinal tract, Kidneys, Liver

Mitigation Suggestions:

- Use of proton pump inhibitors
- Allergy screening
- Regular renal function monitoring

Pharmacokinetics

Absorption:

Oral bioavailability of approximately 90%

Distribution:

Widely distributed with moderate protein binding

Metabolism:

Primarily hepatic via CYP450 enzymes

Excretion:

Renal excretion as unchanged drug

Half-life: 1-2 hours

Detailed Analysis

Amoxylirin combines the beta-lactam antibacterial effects of amoxicillin with the anti-inflammatory action of aspirin, targeting essential bacterial cell wall synthesis machinery as well as the cyclooxygenase pathway involved in inflammation and pain. This dual mechanism aids in treating infections with a significant inflammatory component, while presenting moderate risks of gastrointestinal complications and allergy. Through innovative structure-based design and detailed pharmacokinetic analysis, Amoxylirin seeks to simplify and enhance therapeutic regimens, potentially reducing the complexity of care and improving patient adherence. Clinical translation focuses on effectiveness and long-term safety, particularly in patients requiring continued anti-inflammatory management.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>